

Standard Form — Writing and Converting

AQA · KS4 Year 9–10

Answer each question in your workbook. Show your working.

VISUAL EXAMPLE

$$5\,600\,000 = 5.6 \times 10^6$$

move the point 6 places left

Standard form: $A \times 10^n$ with $1 \leq A < 10$.

COMMON MISTAKES

- ✗ A must satisfy $1 \leq A < 10$. So 36×10^4 is wrong; correct: 3.6×10^5 .
- ✗ Positive power = big; negative power = small.
- ✗ Count places carefully — 0.00047 needs 4 moves $\rightarrow 4.7 \times 10^{-4}$.
- ✗ $0.003 = 3 \times 10^{-3}$, NOT 3×10^3 .

METHOD

- 1 Move decimal so $1 \leq A < 10$
- 2 Count places moved = power n
- 3 Big number (≥ 1) $\rightarrow$ positive n
- 4 Small number (< 1) $\rightarrow$ negative n
- 5 Check by converting back

WORKED EXAMPLE

Write 0.000 47 in standard form.

Step 1 — Move decimal 4 places: 4.7

Step 2 — Original < 1 , so $n = -4$

Answer: 4.7×10^{-4}

